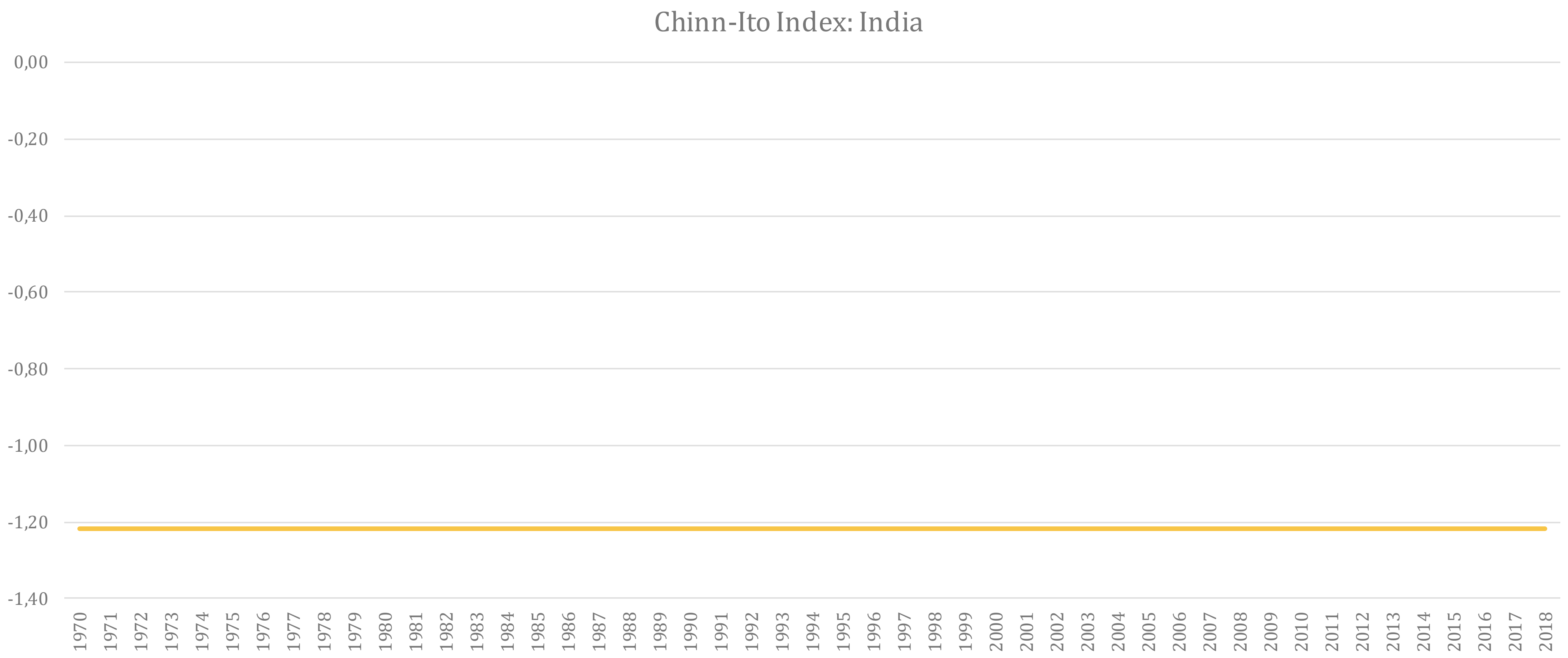
CAL in India

Comment by Josh Felman
EMF 2020

Summary of Paper

- Question: How free is India's capital account?
- Traditional research approach: look at restrictions!
- Problem 1: *De jure* not the same as *de facto*: there are loopholes
- Problem 2: Not all restrictions have the same importance: how to net?
  - 2008: 11 easing measures, 4 tightening
  - 2018: 14 easing, 9 tightening

Standard measure: No liberalization?



Paper's approach

- Logic
  - Brilliantly simple!
  - Liberalization means greater linkages between domestic and foreign financial markets
  - So: test to see how well arbitrage is working

- Specific test: Covered Interest Parity (CIP)
  - Domestic interest rates = foreign rates + expected currency depreciation
  - Expected depreciation = forward rate / spot rate
  - **Arbitrage will ensure equation holds**

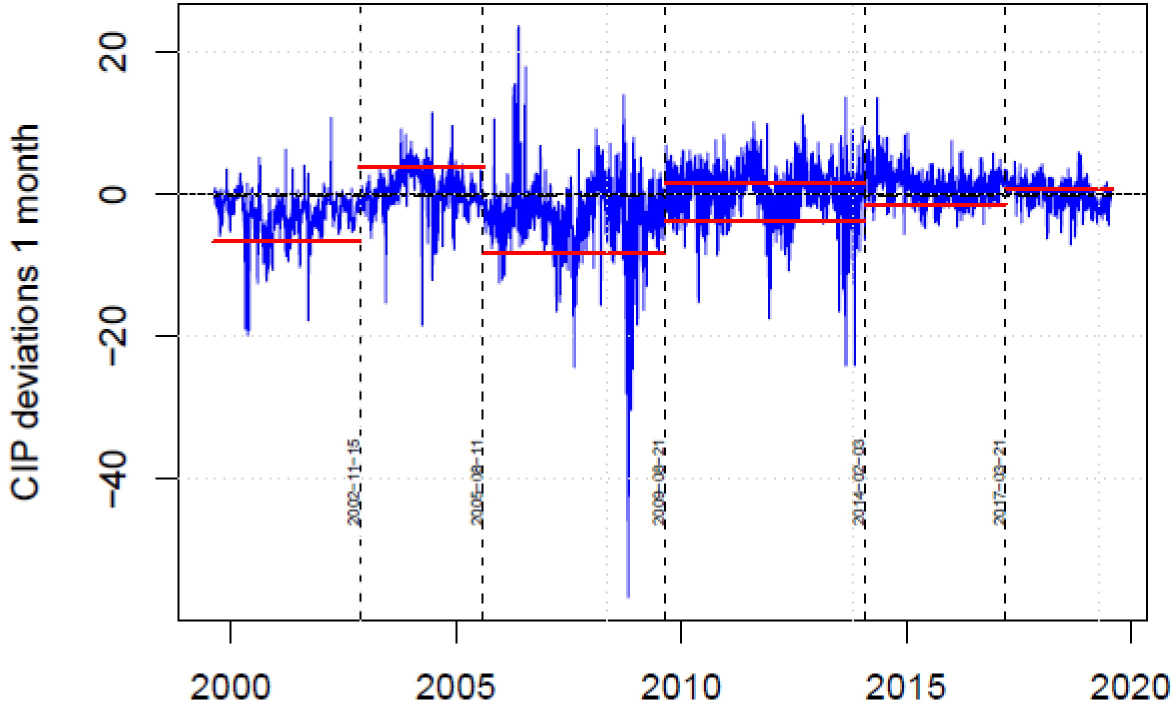
Findings

- Finding 1: CIP does not hold, restrictions have some impact
- Finding 2: But deviations have diminished sharply in past decade
- Implication: capital account has become much freer

Key chart

- No-arbitrage bands diminish

- Change: after 2010



Why does this matter?

- Critical input for monetary policy!

- Case 1: arbitrage is perfect
  - Domestic interest rate = foreign rate + expected depreciation
  - Pegging rate ⟹ expected depreciation is 0
  - So domestic interest rates cannot deviate from foreign rates
  - Implication: RBI must choose between setting interest rate and targetting exchange rate (the "trilemma")

- Case 2: arbitrage is imperfect
  - Implication: RBI can target both, but will be costly
  - Intervention in foreign exchange market increases domestic money supply, fueling inflation
  - So needs to be "sterilized" by selling bonds

How costly?

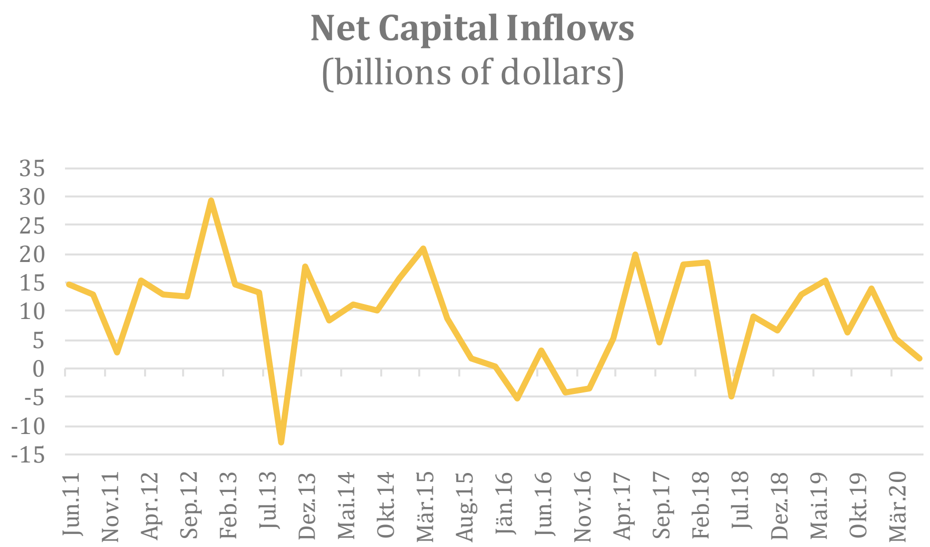
- Assume
  - Domestic interest rates 6 percent, foreign rates 0 percent
- Case I: increasing interest rates attracts $10 billion in arbitrage
  - Annual cost of sterilization will be $600 million
- Case II: attracts $100 billion
  - Annual cost: **$6 billion!**

Implications

- Clearly RBI will be much more reluctant to target exchange rate (or raise rates) in case II
- Not theoretical case: in 2007, capital inflows reached 9 percent of GDP!
- So **precise results of paper** are critical for monetary policy formulation

How reliable are the results?

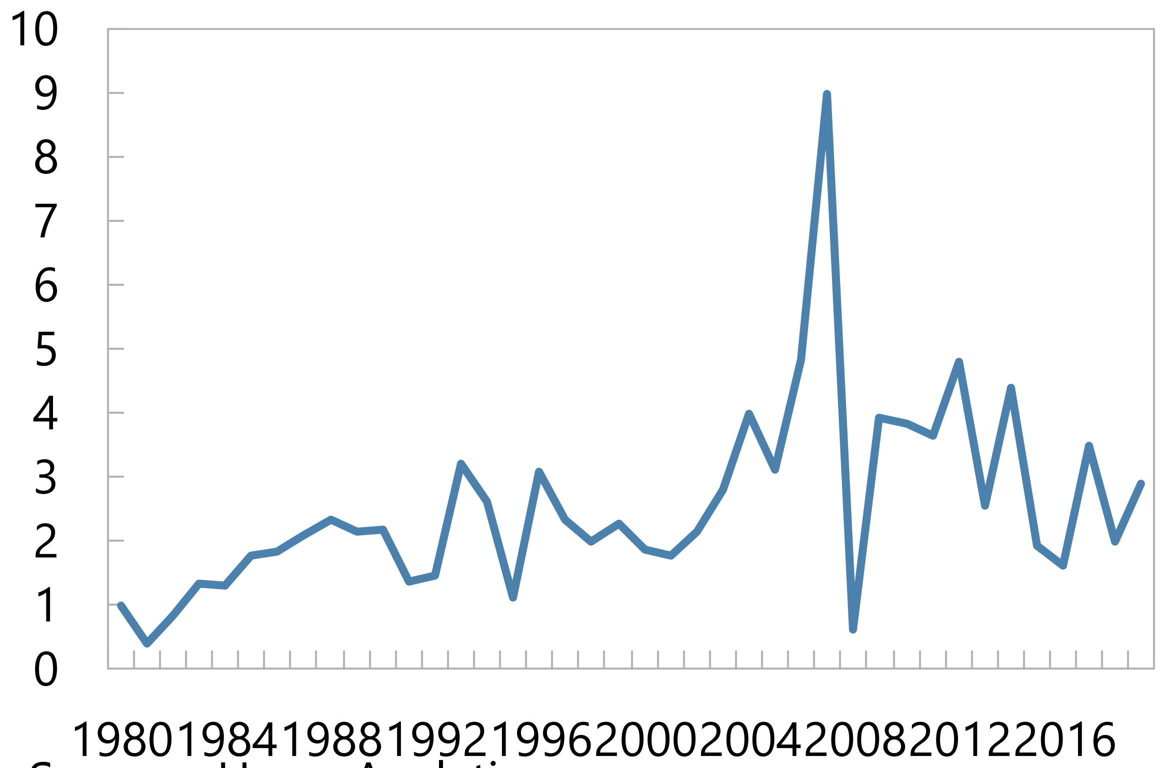
- Cross-check: price/quantity
- CAL ⇨ more capital inflows
- But there's no trend



How reliable are the results/2

- Price measure: CAL in 2010s
- Quantity: CAL in early 2000s

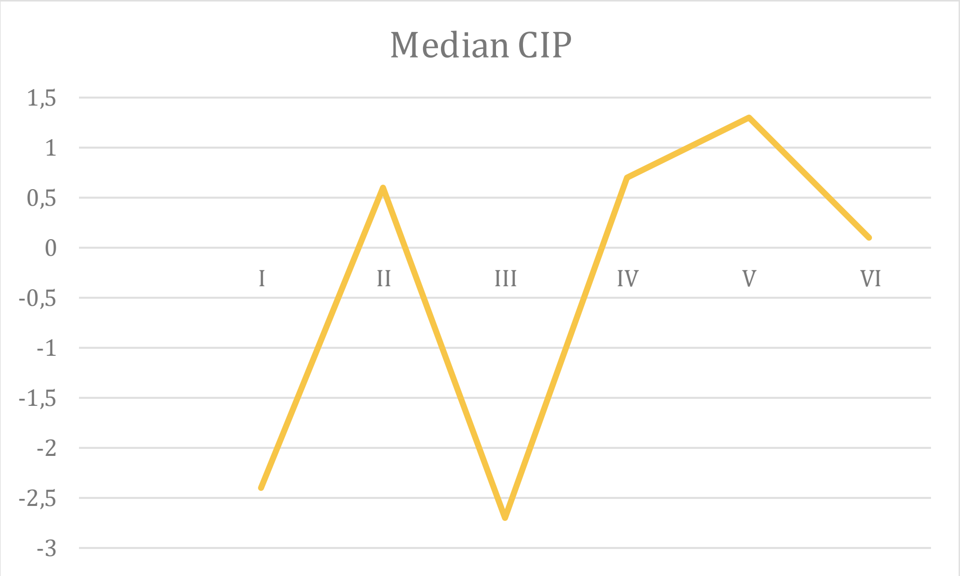
India: Net Capital Inflows
Percent of GDP



Sources: Haver Analytics

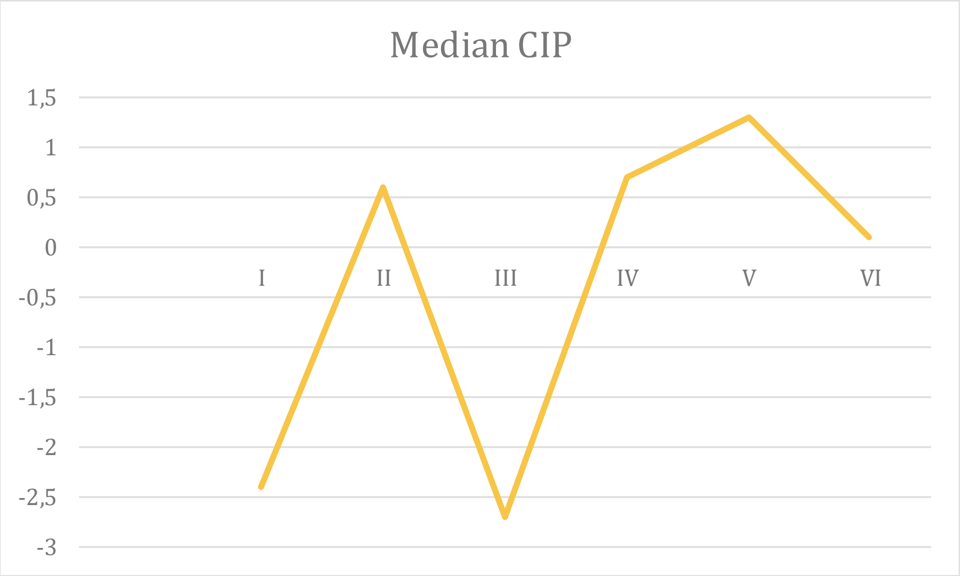
How reliable are results/3

- Examine CIP deviations directly
- Paper finds six periods
- Is there really a trend?



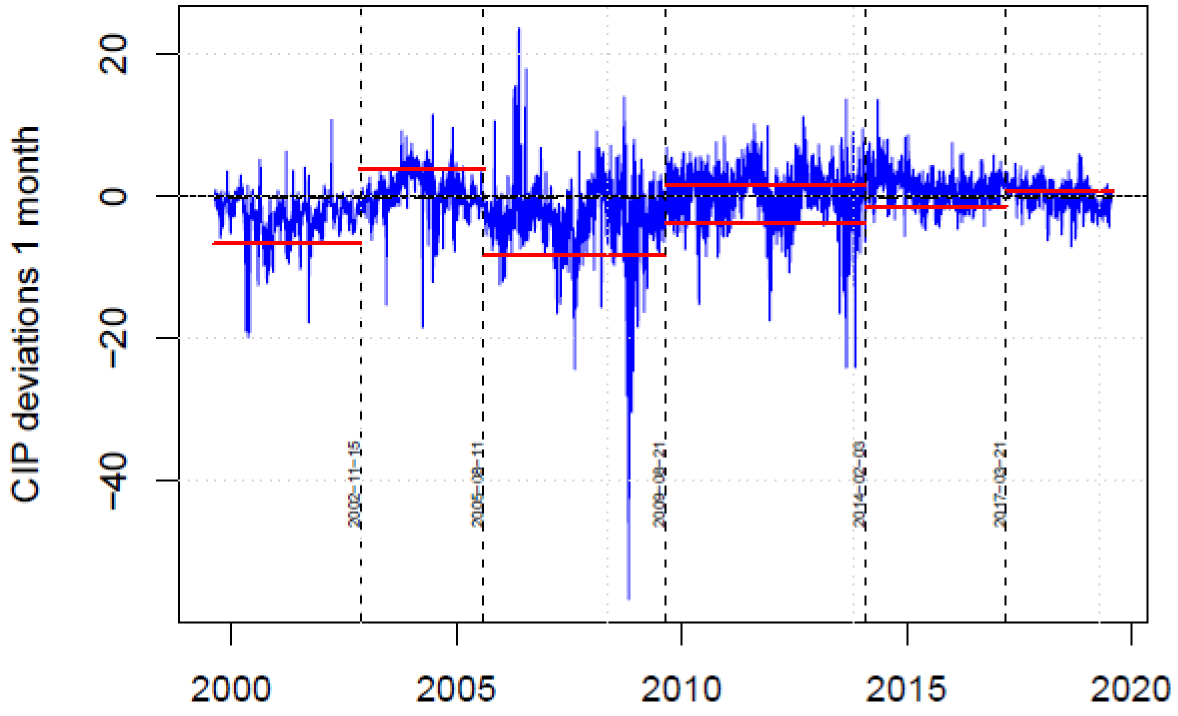
Explanations?

- Period III spike
  - Paper notes 4 tightening measures
  - But there were 11 loosening measures
- Period V spike
  - No explanation
- Period VI reduction
  - Paper cites 14 easing measures
  - But there were 9 tightening
- Inference
  - Circular reasoning?
  - Reverse causality?



Further work: robustness checks

- Econometrically robust?



Further work: price/quantity cross-checks

- Do CIP deviations correlate with capital flows?
  - Trends
  - Short term event studies
- Are flows smaller when CIP deviations are in no-arbitrage bands?
- If not, why not?

Further work: price/regulation cross-checks

- Do (certain) tightening regulations lead to higher CIP deviations?

- Do loosening regulations reduce deviations?

- Problem: causality goes both ways
  - Solution: event studies

- Can we use CIP to assess significance of regulatory moves?

Further work: add context

- How does India's CAL compare to other countries?
- CIP deviations in other BRICs
- Size of no-arbitrage bands in BRICs

Conclusion

- Important paper
- Exceptionally clearly written
- Interesting preliminary results
- Look forward to the final version!

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